Section II Study Revision: Models, Baseline, and Interface

Working proposal — September 5, 2026

This standalone review preserves the evaluated controller and models. It proposes a more explicit Section II and an editable replacement for Figure 1; it is not the final ACC submission layout.

II. Simulation Models, Attitude Baseline, and Actuator Interface

A. MuJoCo Platforms and CF2 Actuation Extension

We use Crazyflie 2.0 (CF2) and Skydio X2 models derived from MuJoCo Menagerie [1]. Each vehicle has six rigid-body degrees of freedom: three translations and three rotations, represented by seven free-joint position coordinates (position and unit quaternion) and six velocity coordinates. The attitude controller requests three body torques; a separate classical height loop requests collective thrust. Thus attitude tracking is evaluated in a freely translating model, not an attitude-only joint model.

The upstream CF2 interface provides collective thrust and three abstract body-moment channels; its documentation identifies the original actuator limits as arbitrary and requiring tuning [2]. Our four-rotor actuation extension replaces that interface with four nonnegative, spatially located thrust inputs while retaining the source mass, inertia, and visual geometry. Rotor moment arms and signed reaction-torque coefficients define an explicit allocation matrix, making the coupling between collective thrust, attitude torque, and individual actuator limits testable. This is a model-interface contribution, not a complete aerodynamic identification of CF2.

TABLE I
Compiled simulation-model characteristics. CF2 authority is scaled relative to the upstream model limit, not a manufacturer rating. Hardware comparisons are given in Table II.

Property	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Unscaled rotor limit (N)	0.0875	13
Main-test authority scale	1.5	1
Main-test rotor limit (N)	0.13125	13
Main-test total limit (N)	0.525	52
Maximum thrust/weight	$1.321 \rightarrow 1.982$	4.001
Rigid-body integrator	RK4	Euler
Experiment timestep (s)	0.0005	0.0005
Motor activation/RPM states	None	None

For CF2, the unscaled model's total thrust limit is 0.35 N. We evaluate a rotor-interface model with increased actuator authority: multiplying $f_{\max,i}$ by 1.5 yields 0.525 N total (Table I). This raises the modeled maximum thrust/weight from 1.321 to 1.982 at $g = 9.81$ m/s^2. For physical perspective, constant-height flight at a single-axis 45° tilt requires $T/(Mg) = 1/\cos 45^\circ = \sqrt{2}$ even before reserving differential-thrust margin. The unscaled model cannot meet that idealized requirement; this does not establish such a limitation for stock CF2 hardware. The scale changes a simulation thrust bound, not rotor speed, stock propulsion capability, or tracking performance by 50%. Mass, inertia, and reaction coefficients are unchanged. PD and CMG use the same model and actuator bounds; any benefit from increased model authority is therefore distinct from the benefit of neural augmentation.

TABLE II
Simulation parameters versus manufacturer information. "Not reported" refers to the manufacturer sources reviewed, not proof that the parameter is unavailable elsewhere.

Quantity	Simulation	Manufacturer information
CF2 mass	27 g	27 g [3]
X2 mass	1325 g	1325 g with battery [4]
CF2 total thrust	$0.350 \rightarrow 0.525$ N	0.568 N measured at 93.75% PWM[a] [5]
X2 rotor limit	13 N per rotor	Not reported [4]
CF2 motor response	No motor dynamics	$\sim$ 180 ms: $0 \rightarrow 25{,}000$ RPM[b] [6]
Inertia tensors	CF2 inherited; X2 geometry-derived	Not reported in product specifications [3], [4]

[a]Historical total-thrust measurement with a power-supply test setup, not a guaranteed continuous battery-powered rating. [b]One motor's reported full-command response with a charged battery; not an identified first-order time constant.

The masses agree with published specifications, but this does not validate all dynamic parameters (Table II). Bitcraze's historical measurement of 57.9 gram-force corresponds to approximately 0.568 N; the evaluated 0.525 N limit is about 7.5% below that operating point [5]. This supports a plausible thrust scale, not hardware validation, and does not establish that a stronger-than-stock motor is required. CF2 inertia is inherited from Menagerie, which cites system-identification work [2]; the rotor-extension moment arms and $|\kappa_i| = 0.005$ m remain design assumptions. X2 inertia, 13 N rotor limits, and $|\kappa_i| = 0.0201$ m are simulation parameters, not confirmed manufacturer characteristics.

Geometric dimensions also require like-for-like definitions. Bitcraze lists $92 \times 92 \times 29$ mm motor-to-motor/height dimensions [3]; our sites give an opposite-motor separation of 91.92 mm. Skydio lists an unfolded flying envelope of approximately

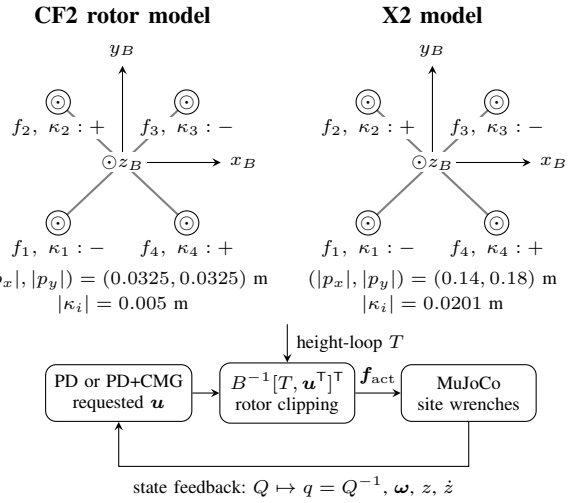


Fig. 1. Code-aligned actuation and feedback interface. Top: body-axis projections, drawn at independent scales, with rotor numbering matching the mixer columns. Circles with dots indicate $+z_B$ thrust; $+/-$ labels give reaction-torque signs, not simulated rotor speeds. X2 site heights are 0.05 m for rotors 1–2 and 0.08 m for 3–4; CF2 sites have zero height. Bottom: both controllers use the same thrust allocation, clipping, and state-feedback interface. Table I specifies model authority, not a manufacturer rating; Table II provides the hardware comparison. (14) distinguishes requested from realized wrench.

$663\times569\times211$ mm [4], whereas the model's rotor centers span 280×360 mm. Center spacing and overall flying envelope are different quantities and do not by themselves establish geometric agreement or mismatch. Propulsion identification, motor and battery dynamics, and experimental validation remain necessary for hardware transfer; no endurance or electrical-power fidelity is claimed.

MuJoCo [7] integrates rigid-body motion, gravity, site-applied actuator wrenches, and enabled passive/contact forces. Both models specify air density 1.225 kg/m^3 and viscosity 1.8×10^{-5} Pa s; their default inertia-based fluid model [8] is a coarse drag approximation, not rotor-resolved aerodynamics. The runner overrides the timestep but retains each model's integrator. Actuators apply algebraic thrust commands without motor activation dynamics. Although gyro, accelerometer, and orientation sensors are defined, the evaluated feedback is read directly from `qpos`/`qvel`; optional measurement perturbations are separate runner settings. These experiments do not establish performance with a hardware state estimator.

B. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body coordinates to inertial coordinates, and let $\boldsymbol{\omega}$ be body angular velocity. With physical inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes, the free-flight rotational balance is

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\text{act}} + \boldsymbol{\tau}_{\text{p}} + \boldsymbol{\tau}_{\text{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\text{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\text{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench. The requested torque and actuator torque are distinct signals in Fig. 1.

The controller conjugates MuJoCo's orientation and uses the passive quaternion $q = Q^{-1}$, so

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired q_d uses the same convention. Define

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \operatorname{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \operatorname{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at $q_{e,0} = 0$; it does not establish robust hybrid global stability [9]. For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T}\boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2}\boldsymbol{q}_{e,v}^\mathsf{T}\boldsymbol{\omega}_e. \tag{8}$$

This is a kinematic identity, not a convergence result. Reference Euler triples parameterize q_d in the passive convention, rather than active body-to-world Euler commands.

C. Physical Inertia and Implemented PD Gains

MuJoCo stores principal moments $\boldsymbol{j}_{\text{p}}$ in its inertial frame [10]. If R_{BI} maps that frame into body axes, the physical inertia is

$$\boldsymbol{J} = R_{BI}\,\operatorname{diag}(\boldsymbol{j}_{\text{p}})R_{BI}^\mathsf{T}. \tag{9}$$

The evaluated implementation uses a tuning vector $\boldsymbol{j}_{\text{tune}} = \boldsymbol{j}_{\text{p}}$ directly, in the returned array order:

$$\boldsymbol{j}_{\text{tune}}^{\text{CF2}} = (2.3951, 2.3951, 3.2347)^\mathsf{T}10^{-5},$$
$$\boldsymbol{j}_{\text{tune}}^{\text{X2}} = (0.0607113, 0.0364684, 0.0254117)^\mathsf{T},$$

in kg m^2. CF2's inertial axes coincide with its body axes. For X2 they do not; its physical body-frame tensor is approximately

$$\boldsymbol{J}_{\text{X2}} = \begin{bmatrix} 0.0366517 & 0 & -0.0021000 \\ 0 & 0.0254117 & 0 \\ -0.0021000 & 0 & 0.0605280 \end{bmatrix} \text{ kg m}^2. \tag{10}$$

Thus the tuning vector must not be identified with $\operatorname{diag}(\boldsymbol{J})$ for both vehicles.

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\text{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta\boldsymbol{j}_{\text{tune}} \odot \boldsymbol{\omega}_n, \tag{11}$$

$$\boldsymbol{u}_0 = \operatorname{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{12}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^\mathsf{T}$ rad/s, $\zeta = 0.95$, and $\operatorname{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. For these rotor-interface models, the model-policy torque limits are $\ell_i = \max(j_{\text{tune},i}a_i, 10^{-6}\ \text{N m})$, with $\boldsymbol{a} = (12, 12, 10)^\mathsf{T}$ rad/s^2. These are controller torque bounds, not guaranteed feasible rotor torques. The law is quaternion

feedback [11]; $\boldsymbol{\omega}_n$ parameterizes its gains but is not a verified set of closed-loop natural frequencies for coupled X2 dynamics. Desired-rate feedforward and gyroscopic cancellation are omitted. The identical fixed gains are retained for PD and its CMG augmentation.

For perspective only, take a constant reference, scalar proportional gain $\boldsymbol{k}_q = k_q\mathbf{1}$, the ideal plant ($\boldsymbol{\tau}_{\mathrm{p}} = \boldsymbol{\tau}_{\mathrm{d}} = 0$), no saturation or residual, and a trajectory confined to one smooth sign branch. Then $V = \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{\omega} + 2k_q(1 - |q_{e,0}|)$ gives $\dot{V} = -\boldsymbol{\omega}^{\mathsf{T}}\operatorname{diag}(\boldsymbol{k}_\omega)\boldsymbol{\omega}$. This restricted identity does not prove stability of the anisotropic, saturated, moving-reference experiments.

D. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{13}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^{\mathsf{T}}$ and the actuator's local reaction torque $\kappa_i f_i$. Both models have zero body-frame center-of-mass x, y offsets, so the site's x, y coordinates also give the relevant center-of-mass moment arms. Figure 1 fixes the rotor ordering used by B; κ_i has units of meters, not angular velocity. The runner computes

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \operatorname{clip}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B\boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{14}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}} \in [\mathbf{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [12].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \operatorname{clip}_{[0, 2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)}\right). \tag{15}$$

This expression uses the model's $g = 9.81$ m/s^2. The height loop reads free-joint height and vertical velocity; the denominator compensates tilt with a lower guard. Its upper bound does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

References

[1] Google DeepMind, "MuJoCo Menagerie," https://github.com/google-deepmind/mujoco_menagerie, model repository; accessed September 5, 2026.

[2] ——, "Bitcraze Crazyflie 2 Description (MJCF)," https://github.com/google-deepmind/mujoco_menagerie/blob/main/bitcraze_crazyflie_2/README.md, model provenance and actuator-limit caveat; accessed September 5, 2026.

[3] Bitcraze AB, "Crazyflie 2.0: Mechanical and Flight Specifications," https://store.bitcraze.io/products/crazyflie-2-0, accessed September 5, 2026.

[4] Skydio, "Skydio X2E Datasheet," https://pages.skydio.com/rs/784-TUF-591/images/skydio-x2e-datasheet-x2-pg.pdf, aircraft specifications; accessed September 5, 2026.

[5] Bitcraze AB, "PWM to Thrust: Old Measurements (2015)," https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/functional-areas/pwm-to-thrust/, historical test setup and total-thrust table; accessed September 5, 2026.

[6] ——, "Measuring Propeller RPM: Part 3," https://www.bitcraze.io/2015/02/measuring-propeller-rpm-part-3/, Feb. 2015, published February 9, 2015; accessed September 5, 2026.

[7] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[8] MuJoCo Developers, "MuJoCo Fluid Forces: Inertia Model," https://mujoco.readthedocs.io/en/3.1.1/computation/fluid.html, accessed September 5, 2026.

[9] C. G. Mayhew, R. G. Sanfelice, and A. R. Teel, "Quaternion-based hybrid control for robust global attitude tracking," *IEEE Transactions on Automatic Control*, vol. 56, no. 11, pp. 2555–2566, 2011.

[10] MuJoCo Developers, "MuJoCo XML Reference: Body/Inertial," https://mujoco.readthedocs.io/en/stable/XMLreference.html#body-inertial, accessed September 5, 2026.

[11] E. Fresk and G. Nikolakopoulos, "Full quaternion based attitude control for a quadrotor," in *2013 European Control Conference (ECC)*, 2013, pp. 3864–3869.

[12] J. C. Monteiro, F. Lizarralde, and L. Hsu, "Optimal control allocation of quadrotor UAVs subject to actuator constraints," in *2016 American Control Conference (ACC)*, 2016, pp. 500–505.